DB Relations

The system as built on branch `mini-erp` (Version 2.0): the base tables first, then what Purchase, QT 2.0 and User 2.0 added or changed. Sources (Purchase Logic Draft2, QT Upgrade 2.0, User Version 2.0).

Current version summary (as built)

23 tables in 5 groups. Parts 1 and 2 below give every key and rule; this is the short form.

Group	Tables	Main relations
Users and files	User, Attachment	Attachment.uploaded_by -> User. Attachment points to its owner (PurchaseOrder, QtReference or ProductionOrder) by `owner_type` + `owner_id`, with no database key
Master data	Customer, Product, Material, MaterialLot, ProductBom, Supplier, SupplierMaterial	Product -> ProductBom -> Material (a BOM line). Material -> MaterialLot (stock by lot). Supplier -> SupplierMaterial -> Material (what a supplier sells, with its price)
Sale	QtReference, QtLine, ProductionOrder, OrderLine, OrderProcess, ProgressEntry	Customer -> QtReference -> QtLine -> Product. Confirm QT makes one ProductionOrder (QtReference.production_order_id, one to one). Order -> OrderLine -> OrderProcess -> ProgressEntry (by User)
Accounting	Invoice, InvoiceLine, Payment, Receipt	One Invoice per ProductionOrder; one Payment and one Receipt per Invoice
Purchase and stock	PurchaseOrder, PurchaseOrderLine, PurchasePayment, StockChange	Supplier -> PurchaseOrder -> PurchaseOrderLine -> Material. One PurchasePayment per PO. StockChange (material, user, and optionally the order or the PO) is the history of every lot change

Rules to remember:

- **Delete:** children go with their parent (lines, processes, BOM, lots, supplier items, attachments). Anything else that is referenced answers `409 IN_USE`. Only a `Draft` QT or `Draft` PO can be deleted. Invoices, receipts and stock history are never deleted.
- **One to one:** Invoice per order, Payment and Receipt per invoice, PurchasePayment per PO, ProductionOrder per confirmed QT.
- **Unique:** User.email, and the codes and numbers of Product, Material, Supplier, QT, order, invoice, receipt and PO; (material, lot number); (supplier, material); (QT, product).
- **Status:** QT `Draft` -> `Confirmed`. Order `Draft` -> `In Progress` -> `Completed` (computed). Invoice `Unpaid` -> `Paid`. PO `Draft` -> `Sent` -> `Received`.
- **Stock moves** only in StockChange + MaterialLot: Start production deducts the BOM (oldest lot first), Receive PO adds once, and a person can Add, Deduct or Adjust a lot by hand.

Part 1: The base tables (the Mini ERP before Version 2.0)

Table-by-table list of keys, relations and delete rules. The whole model, as built, is in the ER Diagram. The rows of QtReference, Material and User are changed by Part 2 below. Source: docs/api (`master-data.md`, `stock.md`, `qt.md`, `orders.md`, `invoices.md`, `receipts.md`).

Foreign keys

Table	Column	References	Cardinality	Delete rule
QtReference	customer_id	Customer.id	many to one	customer in use -> `409 IN_USE`
ProductionOrder	qt_id	QtReference.id	many to one	QT used by an order -> `409 IN_USE`
ProductionOrder	customer_id	Customer.id	many to one	customer in use -> `409 IN_USE`
OrderLine	order_id	ProductionOrder.id	many to one	goes with the order
OrderLine	product_id	Product.id	many to one	product in use -> `409 IN_USE`
OrderProcess	line_id	OrderLine.id	many to one	goes with the line
ProgressEntry	process_id	OrderProcess.id	many to one	never deleted
ProgressEntry	user_id	User.id	many to one	user with history -> `409 IN_USE`
ProductBom	product_id	Product.id	many to one	goes with the product
ProductBom	material_id	Material.id	many to one	material in a BOM -> `409 IN_USE`
MaterialLot	material_id	Material.id	many to one	goes with the material
StockChange	material_id	Material.id	many to one	material with a non-initial change cannot be deleted
StockChange	order_id	ProductionOrder.id	many to one, optional	set when the start of an order deducts stock
StockChange	user_id	User.id	many to one	never deleted
Invoice	order_id	ProductionOrder.id	one to one	one invoice per order
InvoiceLine	invoice_id	Invoice.id	many to one	goes with the invoice
Payment	invoice_id	Invoice.id	one to one	one full payment per invoice
Receipt	invoice_id	Invoice.id	one to one	one receipt per invoice

Unique keys

Table	Unique
User	email (lower-cased)
Product	code
Material	code
MaterialLot	(material_id, lot_number)
QtReference	qt_number
ProductionOrder	order_number (`ORD-yyyy-0001`, given by the server)
Invoice	invoice_number, order_id
Payment	invoice_id
Receipt	receipt_number, invoice_id

Lifecycle rules that touch several tables

Event	Tables written	Rule
Start production (`POST /orders/:id/start`)	ProductionOrder.started_at, MaterialLot, StockChange	Deducts the BOM of every line, oldest lot first. `409 STOCK_SHORT` if a material is short. Locks products and processes
Progress entry	ProgressEntry, OrderProcess	Updates the status and completed quantity of the process. An order is Completed when all processes are
Issue invoice	Invoice, InvoiceLine	Only for a Completed order, once. Copies the data (snapshot). Locks the progress of the order
Pay invoice	Payment, Invoice.status, Receipt	Payment date and amount must equal the total (`MUST_EQUAL_TOTAL`). Sets `Paid` and issues one receipt
Manual stock change	MaterialLot, StockChange	`Add`, `Deduct` or `Adjust` a lot at any time. The history is never changed or deleted

Chain of one sale

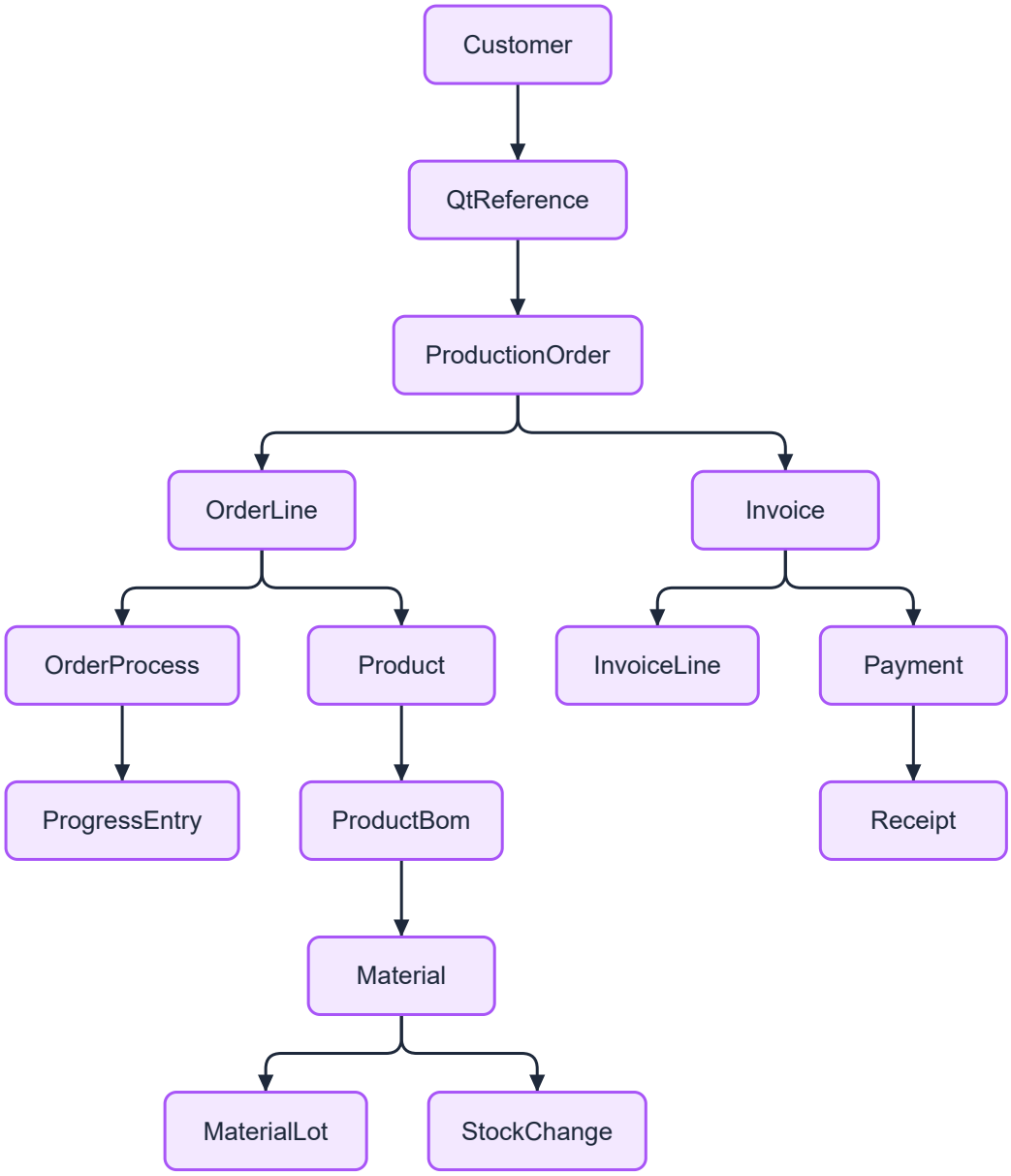


Figure 1. Chain of one sale

Part 2: Added or changed in Version 2.0 (as built)

Keys, relations and rules after the Purchase, QT 2.0 and User 2.0 changes. Only the **new or changed** rows are listed; everything else stays as in Part 1. The order status is `Draft`, `In Progress`, `Completed` (computed; `Draft` replaces `New`). An order is made only by Confirm QT.

New foreign keys

Table	Column	References	Cardinality	Delete rule
SupplierMaterial	supplier_id	Supplier.id	many to one	goes with the supplier

Table	Column	References	Cardinality	Delete rule
SupplierMaterial	material_id	Material.id	many to one	material sold by a supplier -> `409 IN_USE`
PurchaseOrder	supplier_id	Supplier.id	many to one	supplier with a PO -> `409 IN_USE`
PurchaseOrder	created_by	User.id	many to one	user with history -> `409 IN_USE`
PurchaseOrderLine	po_id	PurchaseOrder.id	many to one	goes with the PO; only a `Draft` PO can be deleted
PurchaseOrderLine	material_id	Material.id	many to one	material on a PO -> `409 IN_USE`
PurchasePayment	po_id	PurchaseOrder.id	one to one	one payment per PO
StockChange	purchase_order_id	PurchaseOrder.id	many to one, optional	set when a PO is received
QtLine	qt_id	QtReference.id	many to one	goes with the QT; only a `Draft` QT can be deleted
QtLine	product_id	Product.id	many to one	product on a QT -> `409 IN_USE`
QtReference	production_order_id	ProductionOrder.id	one to one, optional	set when the QT is confirmed
Attachment	uploaded_by	User.id	many to one	never deleted
Attachment	(owner_type, owner_id)	PurchaseOrder, QtReference or ProductionOrder	many to one, **no database key**	goes with the owner (service deletes the file too)

Changed columns

Table	Change
QtReference	adds `status` (`Draft`, `Confirmed`), `customer_po_number`, `customer_po_date`, `production_order_id`; drops `file_name`, `file_type`, `file_path` (moved to Attachment)
Material	adds `unit`
User	rule: `role = Worker` needs `department` in (`Sales`, `Factory`)

Unique keys

Table	Unique
Supplier	code
SupplierMaterial	(supplier_id, material_id)
PurchaseOrder	po_number
PurchasePayment	po_id
QtReference	qt_number, production_order_id

Table	Unique
QtLine	(qt_id, product_id)

Lifecycle rules that touch several tables

Event	Tables written	Rule
Create PO	PurchaseOrder, PurchaseOrderLine	Status `Draft`. Only materials of the chosen supplier (`SupplierMaterial`) can be picked. Unit price starts from `SupplierMaterial.unit_price`, the user can change it. `total` = sum of line amounts
Edit PO	PurchaseOrderLine	Lines change only in `Draft`
Change PO status	PurchaseOrder	`Draft` -> `Sent` -> `Received`, one step forward, by hand
Receive PO	PurchaseOrder.received_at, MaterialLot, StockChange	Adds each line to a lot as a stock change `Add` linked by `purchase_order_id`. **Once only**: a second `Received` does not add again
Pay PO	PurchasePayment	Needs status `Received`. One payment per PO
Upload file	Attachment	Kinds `PO_FILE`, `QUOTATION`, `EVIDENCE`, `OTHER`. PDF, PNG or JPG, up to 10 MB. A new `PO_FILE` replaces the old one
Confirm QT	QtReference.status, ProductionOrder, OrderLine, QtReference.production_order_id	Needs a customer and at least one line. Creates one Draft Factory Order from the QT lines. Once only. The customer PO file is attached to the order. Locks the QT lines
Start production	as before	Unchanged

Chain of the new flows

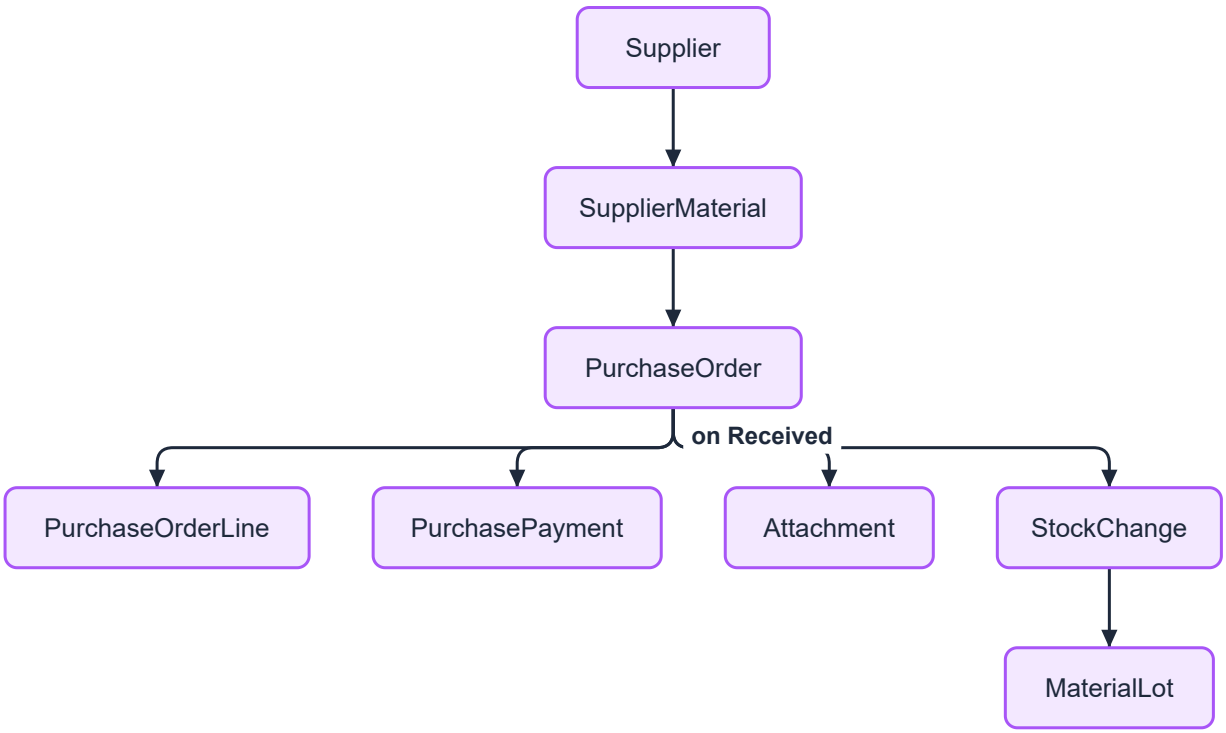


Figure 2. Chain of the new flows: Purchase

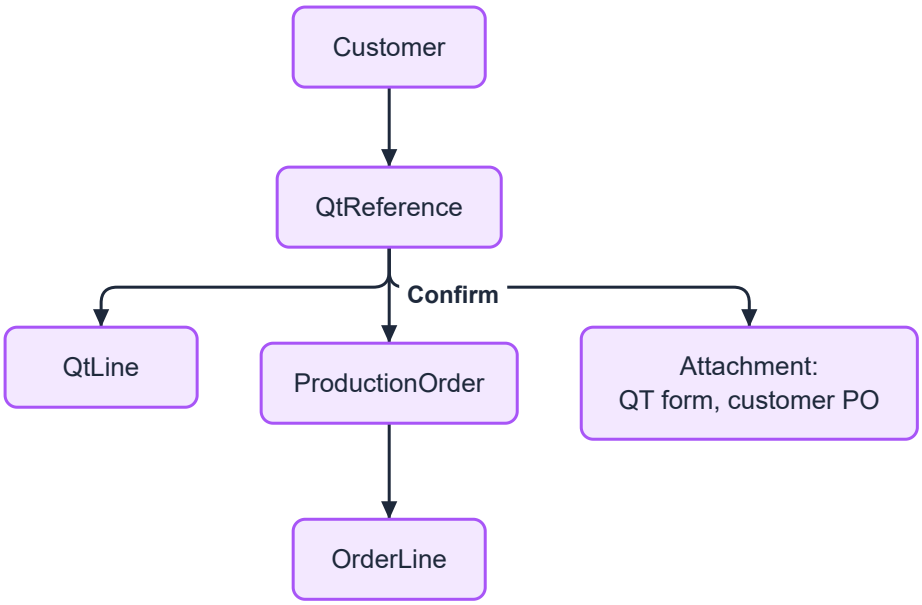


Figure 3. Chain of the new flows: Sale